A B Abhirama

+91 8660120678 | abhiram.b@icloud.com | portfolio.abhiram.tech | github.com/abhiramaab | Bengaluru, India

Summary

Self-taught backend developer building high-reliability services, RESTful APIs, and event-driven pipelines with **Java**, **Spring Boot**, and **PostgreSQL**. Active contributor to core open-source infrastructure (**Apache BifroMQ**, **Spring Cloud**, **Apache Camel**, **Kestra**) spanning authentication, Kafka messaging, and distributed coordination. 4 years of supply chain & ERP operations background with strong systems execution.

Technical Skills

Languages & Frameworks: Java (8/17/21), Spring Boot, Spring Cloud, Spring Security, Spring Data JPA, Hibernate, SQL
Architecture & System Design: Distributed Systems, RESTful APIs, Microservices, Idempotency, Distributed Locks, Transactional Outbox, HLD & LLD
Databases & Caching: PostgreSQL, MySQL, Redis (Caching & Distributed Locking), Database Indexing, Query Optimization
DevOps & Tooling: Docker, Git, GitHub, Maven, AWS (EC2), Linux (Bash), Kafka, Swagger/OpenAPI, Postman, JUnit 5, GitHub Copilot

Open Source Contributions

Merged pull requests into production projects:

Spring Cloud Config | *Server-side config for distributed systems* PR #3272

- Added passwordless Azure Managed Identity authentication to Config Server Git repositories; acquired tokens and injected them as Bearer headers into JGit HTTP transport with clean legacy fallback.

Apache BifroMQ | *High-performance distributed MQTT broker* PR #314 (Merged via #315)

- Diagnosed QoS 1 subscriber stall caused by Netty decoder vs. packet-sizer mismatch (Issue #313); authored fix aligning variable-length integer framing and drop-and-confirm delivery semantics, integrated into main via PR #315.

Camel Kamelets | *Reusable connectors for Apache Camel* PR #3039

- Preserved Kafka message keys across HTTP transport hops in kafka-sink Kamelet, preventing partition key loss and ensuring strict per-key message ordering downstream.

Kestra | *Declarative orchestration engine* #19007 | #51

- Eliminated socket leak in MCP client health-check mechanism occurring on builder instantiations; added case-sensitive enum validation to Telegram plugin parse mode for fail-fast error handling.

Projects

LoomPay | *Java 21, Spring Boot, Redis, PostgreSQL, Docker, System Design* [Demo] | [Code] | [Docs]

- Built payment service with **Idempotency** (double-checked locking) and **Distributed Locks** (Redis SET NX EX) to stop duplicate charges during concurrent payment requests.
- Wired **Transactional Outbox Pattern** with a scheduled worker and retry limits so webhook events never drop when the network fails.
- Designed **Consistent Hashing ring router** with virtual nodes for load distribution, and **Double-Entry Ledger** for financial balance consistency.

Routesphere | *Spring Boot, Spring Cloud, Eureka, Gateway, OpenFeign, Docker* [Demo] | [Code] | [Docs]

- Logistics dispatch app split into 9 Spring Boot services covering orders, dispatch, and tracking.
- Eureka handles service discovery and Spring Cloud Gateway routes traffic with JWT validation done once at the edge.
- Services talk to each other over OpenFeign with client-side load balancing.

Synoptiq | *Java, Spring Boot, Spring Security, PostgreSQL, Docker, AWS EC2* [Demo] | [Docs]

- Workflow automation platform syncing Gmail and Google Calendar via OAuth2 and stateless JWT auth, storing records in PostgreSQL.
- Turned on HTTP/2 multiplexing so concurrent API calls and Server-Sent Events share one persistent TCP connection.
- Implemented background OAuth token refresh lifecycle and exponential backoff retry policies for Google API rate limits.

Experience

Supply Chain & ERP Operations 2021 – 2026
Wipro Consumer Care, Medicon Leather, Mannol India, iThink Logistics *Bengaluru, India*

- Ran SKU catalogs, order pipelines, and SAP ERP inventory reconciliation across 7 marketplaces including Amazon, Flipkart, and Myntra.
- Coordinated warehouse routing and regional fulfillment for a mid-size 3PL operation.

Education

University of Mysore Expected 2028
Bachelor of Computer Applications (BCA), Online *Mysuru, Karnataka*

Rural Institute of Advanced Studies 2018
Higher Secondary Certificate (12th) *Raipur, CG*